Project Cycle 3 Instructions

Two-Sample Inference using `YRBS_2007.csv`

1. Cycle Overview 概要

Project Cycle 3 focuses on **two-sample inference**. In this cycle, each group will develop one statistical research question that compares two groups using the `YRBS_2007.csv` dataset.
本循環重點：**two-sample inference（兩樣本推論）**。
This cycle is more difficult than Cycle 2 because students must now compare **two populations or two groups**, not only estimate one population value. This means the group must pay closer attention to:

- how the two groups are defined;
- whether the response variable is quantitative or categorical;
- which two-sample method is appropriate;
- whether assumptions are reasonable;
- how to interpret the difference between groups.

本循環比 Cycle 2 更困難，因為你們不只是估計單一母體，而是要比較**兩個群體**。

2. Group Structure 小組規定

Project Cycle 3 is completed in groups of **2 students**.
Each group must submit one shared project, but both members must understand the full workflow. During consultation or exhibition, either member may be asked to explain the analysis.
本循環為**兩人一組**。兩位成員都必須理解完整分析流程。

3. Main Task 主要任務

Each group must choose **one** pre-approved research question from Section 5.
Your group must complete:

1. one clear two-sample research question;
2. correct definition of group variable and response variable;
3. appropriate data cleaning and recoding;
4. one suitable two-sample statistical method;
5. confidence interval for the difference, where appropriate;
6. hypothesis test and conclusion;
7. at least two meaningful visualizations;
8. one clear summary table;
9. interpretation in context;
10. one-slide infographic summary for exhibition.

4. Main Statistical Methods 方法範圍

Depending on the chosen research question, the method may include:

- **Two-sample t-test** for comparing two means;
- **Welch two-sample t-test** when equal variance is not assumed;
- **Pooled two-sample t-test** only when equal variance is reasonably justified;
- **Two-proportion z-test** for comparing two proportions;
- **Confidence interval for difference in means**;

- **Confidence interval for difference in proportions**.

Default expectation:

- For two means, use **Welch's t-test** unless there is a clear reason to assume equal variance.
- For two proportions, use a **two-proportion z-test**.

預設原則：**平均數比較優先使用 Welch t-test；比例比較使用 two-proportion z-test**。

5. Pre-Approved Research Questions 預先核准題目

Choose **only one** research question.

Question 1: Gender and Current Cigarette Use

Research question: Is the proportion of current cigarette use different between male and female students?

Variables:

- Group variable: `WhatIsYourSex`
- Response variable: `CurrentCigaretteUse`

Question 2: Gender and Current Alcohol Use

Research question: Is the proportion of current alcohol use different between male and female students?

Variables:

- Group variable: `WhatIsYourSex`
- Response variable: `CurrentAlcoholUse`

Question 3: Gender and Sad or Hopeless Feeling

Research question: Is the proportion of students who felt sad or hopeless different between male and female students?

Variables:

- Group variable: `WhatIsYourSex`
- Response variable: `SadOrHopeless`

Question 4: Gender and Body Weight

Research question: Is the mean body weight different between male and female students?

Variables:

- Group variable: `WhatIsYourSex`
- Response variable: `HowMuchDoYouWeighWithoutShoesInKG`

Question 5: Gender and Height

Research question: Is the mean height different between male and female students?

Variables:

- Group variable: `WhatIsYourSex`
- Response variable: `HowTallAreYouWithoutShoesInMeters`

Question 6: Current Cigarette Use and BMI Percentile

Research question: Is the mean BMI percentile different between students who currently use cigarettes and those who do not?
Variables:

- Group variable: recoded `CurrentCigaretteUse`
- Response variable: `BMIPCT`

Question 7: Current Alcohol Use and BMI Percentile

Research question: Is the mean BMI percentile different between students who currently use alcohol and those who do not?
Variables:

- Group variable: recoded `CurrentAlcoholUse`
- Response variable: `BMIPCT`

Question 8: Sad or Hopeless Feeling and Current Cigarette Use

Research question: Is the proportion of current cigarette use different between students who felt sad or hopeless and those who did not?
Variables:

- Group variable: recoded `SadOrHopeless`
- Response variable: recoded `CurrentCigaretteUse`

6. Required Recoding Rules 重編碼規則

Use the same coding logic from Cycle 2.

Behavior Variables

- `EverCigaretteUse`: success = code 1, failure = code 2
- `SadOrHopeless`: success = code 1, failure = code 2
- `CurrentCigaretteUse`: success = codes 2–7, failure = code 1
- `CurrentAlcoholUse`: success = codes 2–7, failure = code 1

For binary variables, define:

$$1 = \text{success / yes / exposed group}$$

$$0 = \text{failure / no / comparison group}$$

You must document all recoding decisions clearly.

7. What Makes Cycle 3 More Difficult?

Cycle 3 has higher expectations than Cycle 2.

7.1. You must define two groups correctly

In Cycle 2, you estimated one value. In Cycle 3, you compare two groups. The group definition must be clear.
Examples:

- male vs female;
- current cigarette users vs non-users;
- students who felt sad or hopeless vs those who did not.

If the groups are not defined correctly, the statistical conclusion will be unreliable.

7.2. You must choose the correct method

The method depends on the response variable:
- quantitative response $\rightarrow$ compare means;
- binary response $\rightarrow$ compare proportions.

A common mistake is using a mean test for a binary proportion problem, or using a proportion test for a quantitative variable.

7.3. You must interpret the difference

The key result is no longer one mean or one proportion. The key result is a **difference**.
For example:

$$\bar{x}_1 - \bar{x}_2$$

or

$$\hat{p}_1 - \hat{p}_2$$

Your interpretation must explain:
- which group has the higher value;
- how large the difference is;
- whether the difference appears statistically meaningful;
- what the result means in context.

7.4. You must check assumptions more carefully

For two-sample inference, you should consider:
- Are the two groups independent?
- Is the response variable appropriate for the selected method?
- Are sample sizes large enough?
- Are there extreme outliers for quantitative variables?
- Is equal variance assumed or not?

For this cycle, **do not assume equal variance automatically**. Use Welch's test unless you justify pooled variance.

7.5. You must communicate visually

The one-slide summary must show the comparison clearly. A good slide should make the group difference visible.
Recommended visuals:
- bar chart for two proportions;
- boxplot for two means;
- side-by-side histograms;
- confidence interval plot for the group difference.

8. Project Structure and Cookiecutter

Use the same Cookiecutter-style project structure introduced in Cycle 2.

```
project-cycle-3/
   README.md
   data/
      raw/
      processed/
   notebooks/
   outputs/
      figures/
      tables/
      summary/
   report/
   references/
```

The purpose is still organization and reproducibility, not advanced programming.

9. What to Keep in the Project Structure

README.md

Include:
- group number;
- member names;
- selected research question;
- group variable and response variable;
- method used;
- short final conclusion.

data/raw

Keep the original `YRBS_2007.csv` file here.

data/processed

Save cleaned or recoded data if created.

notebooks

Your notebook(s) should show:
- data loading;
- variable selection;
- recoding;
- descriptive comparison;
- two-sample inference;
- interpretation.

outputs

Save final figures and tables.

references

Document:
- variable coding;
- group definition;
- response variable definition;
- method choice;
- assumptions considered.

10. Required Project Components

Your final submission must include:
1. project folder with correct structure;
2. README;
3. variable notes;
4. notebook(s);
5. saved figures;
6. saved summary table;
7. one-slide infographic summary;
8. final interpretation.

11. Required Statistical Output

Your project must report:
- sample size in each group;
- group summaries;
- estimated difference between groups;
- confidence interval for the difference, where appropriate;
- test statistic;
- p-value;
- conclusion at $\alpha = 0.05$;
- interpretation in context.

12. Required Visual Output

Your project must include at least:
- one visualization comparing the two groups;
- one visualization or table showing the inferential result.

Examples:
- bar chart of proportions by group;
- boxplot of quantitative response by group;
- confidence interval plot;
- summary table with group means/proportions and difference.

13. One-Slide Summary Requirement

You must prepare **one single slide** for exhibition.
This must be a **presentation slide**, not a report page. Use PowerPoint, Google Slides, Canva, or similar software.
The slide should work as an **infographic**.
It should include:

- research question;
- group definition;
- response variable;
- method;
- key result;
- conclusion;
- at least one clear figure.

Avoid long paragraphs. The slide should be understandable in about **30–60 seconds**.

14. Consultation Expectations

Cycle 3 includes:

- one project release week;
- one work session without consultation;
- three consultation rounds;
- one exhibition week.

Because there are more groups in Cycle 3, consultation is distributed across several weeks.
By the consultation round, your group should have:

- selected one approved research question;
- defined the two groups;
- cleaned or recoded necessary variables;
- completed descriptive summaries;
- started the main two-sample inference;
- prepared specific questions for feedback.

No draft, no consultation mark.

15. What Will Be Assessed

Cycle 3 is more advanced than Cycle 2. Assessment will focus on:

- correctness of group definition;
- correctness of response variable definition;
- correct method selection;
- correct two-sample inference;
- interpretation of difference between groups;
- assumption awareness;
- quality of figures and tables;
- reproducibility of project structure;
- clarity of one-slide summary;
- ability of both members to explain the work.

16. Common Mistakes to Avoid

- Comparing groups without clearly defining the groups.
- Treating coded categories as continuous numbers.
- Using a two-sample t-test for a binary response variable.
- Using a two-proportion test for a quantitative response variable.
- Reporting a p-value without explaining the direction or size of the difference.
- Claiming causality from observational survey data.
- Ignoring missing or invalid codes.
- Creating a slide that is mostly text.

17. Practical Advice

- Start by identifying the two groups.
- Then identify whether the response is quantitative or binary.
- Choose the method only after the variable types are clear.
- Always report group summaries before test results.
- Interpret the difference, not only the p-value.
- Use the slide to show the comparison visually.
- Make sure both members can explain the full workflow.